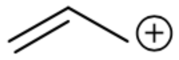
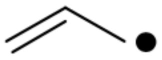
QX4001 – Molecules to Materials

Rotation 3 – Question Set 1

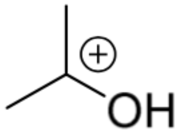
1) For the following species, show the structural formula (show all the carbons, hydrogens and heteroatoms), determine the hybridisation for all the carbons, and write resonance forms when suitable.



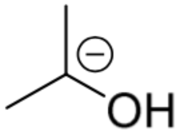
A



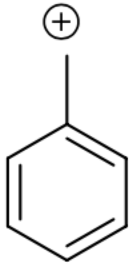
B



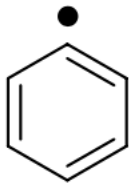
C



D



E



F

1) For the following species, show the structural formula (show all the carbons, hydrogens and heteroatoms), determine the hybridisation for all the carbons, and write resonance forms when suitable.

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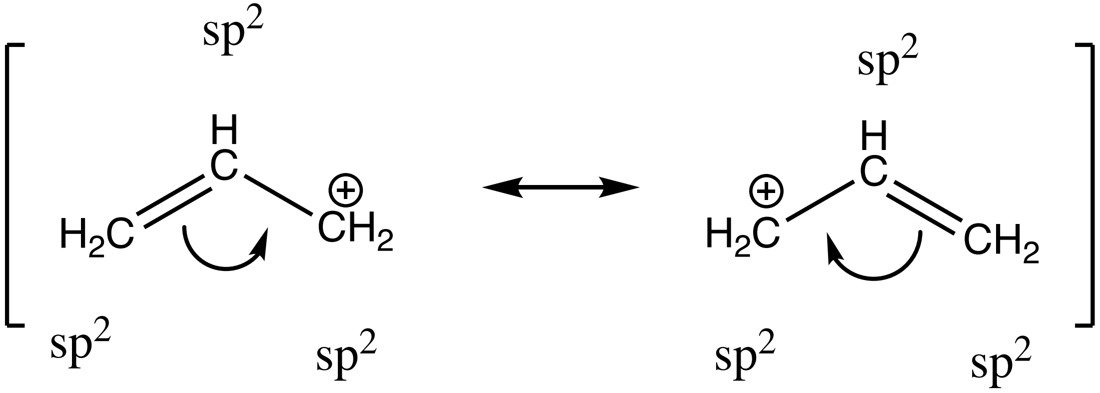
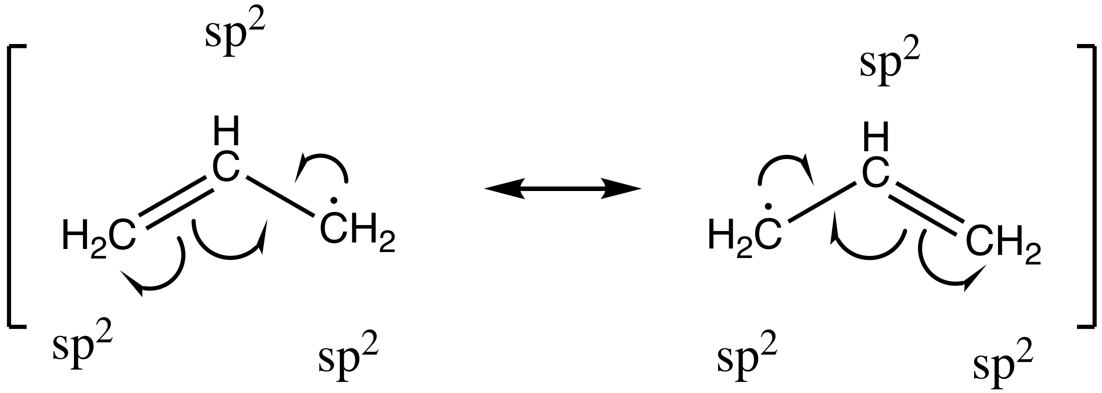


B



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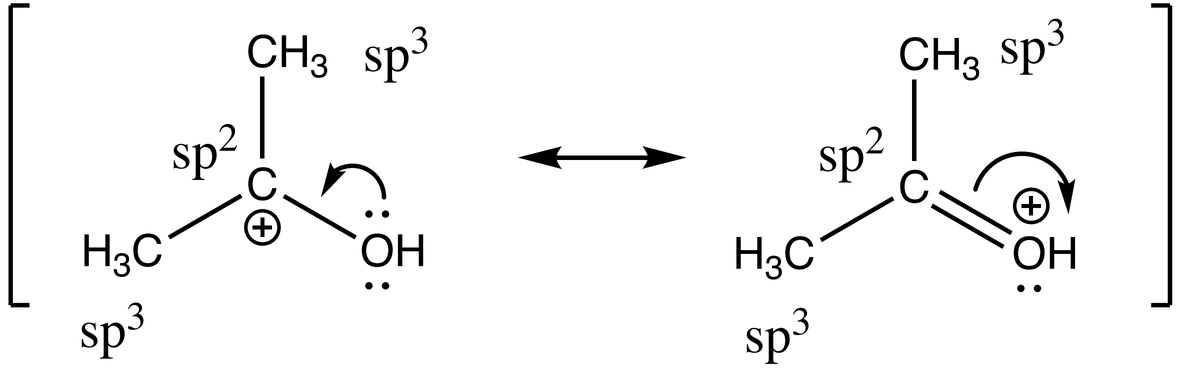

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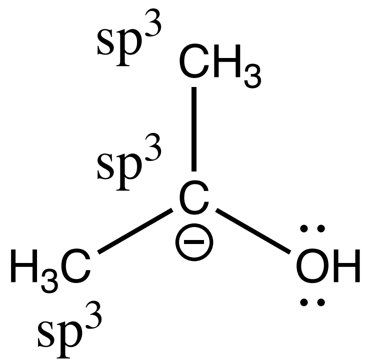

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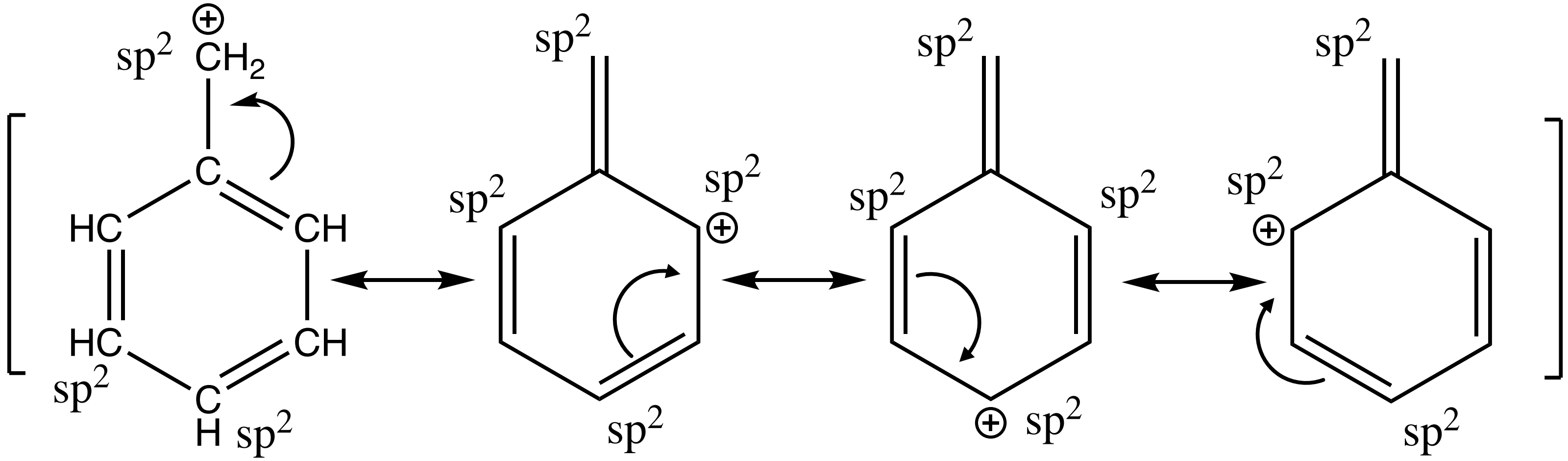
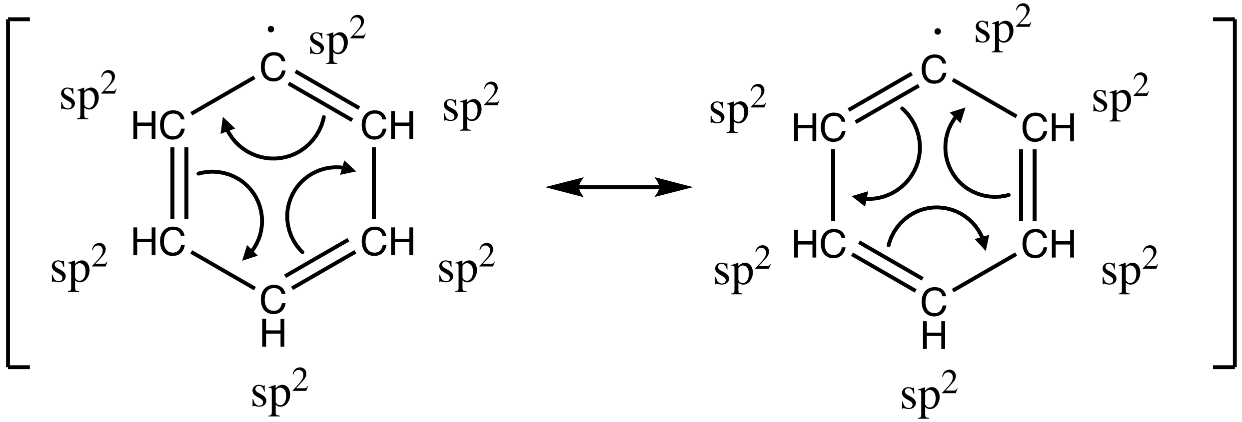
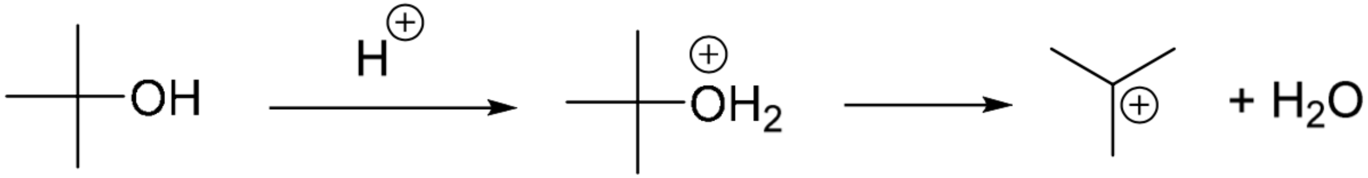


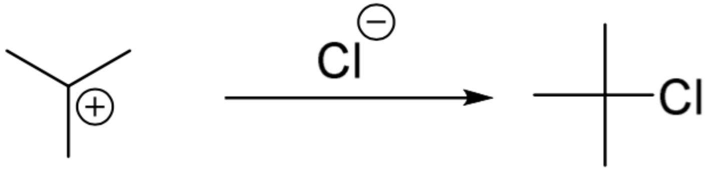
F

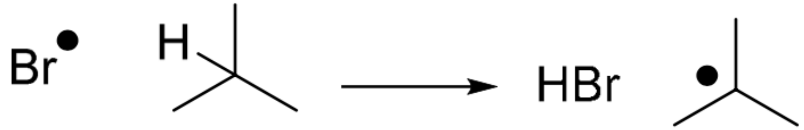


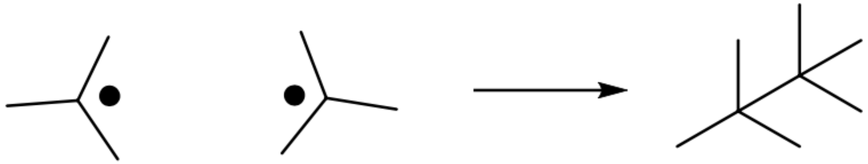
2) Use curly arrows, and lone pairs where appropriate, to complete the following diagrams:

A) $Br{-}Br \xrightarrow{h\nu} 2\ Br^{\bullet}$

B) 

C) 

D) 

E) 

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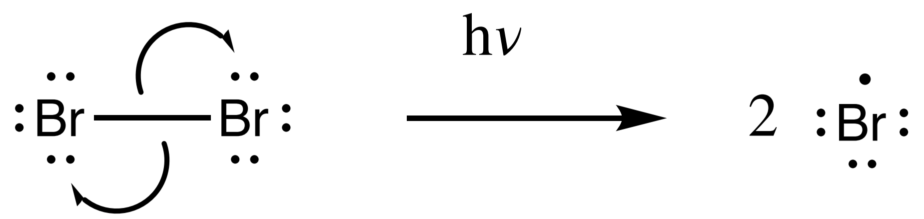
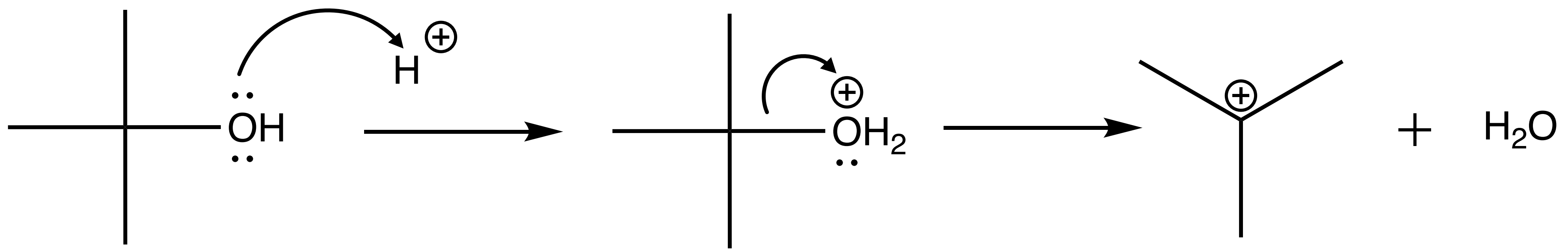


B



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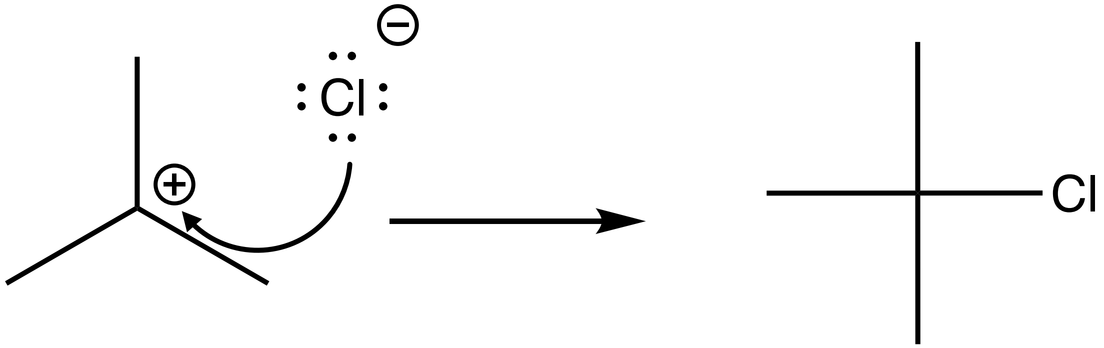
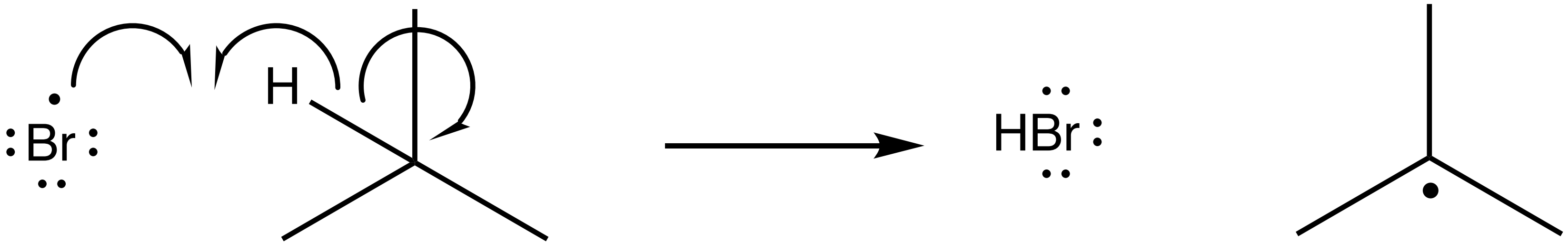


D



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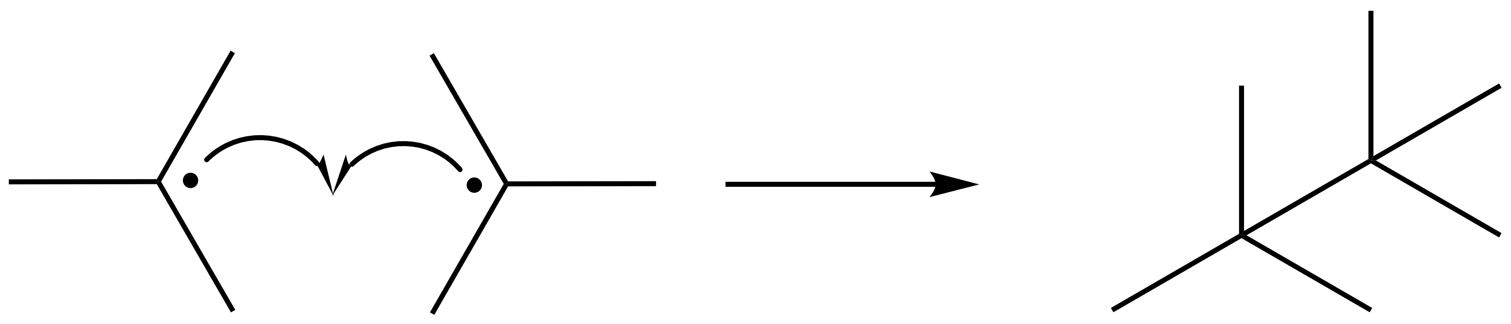

3) Why does the stability of free-radicals follow the trend: tertiary > (more stable than) secondary > primary > methyl?

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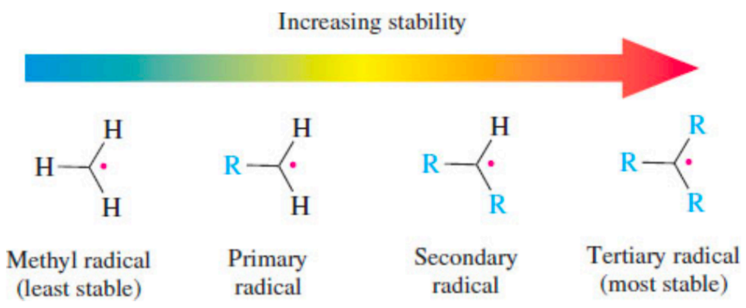
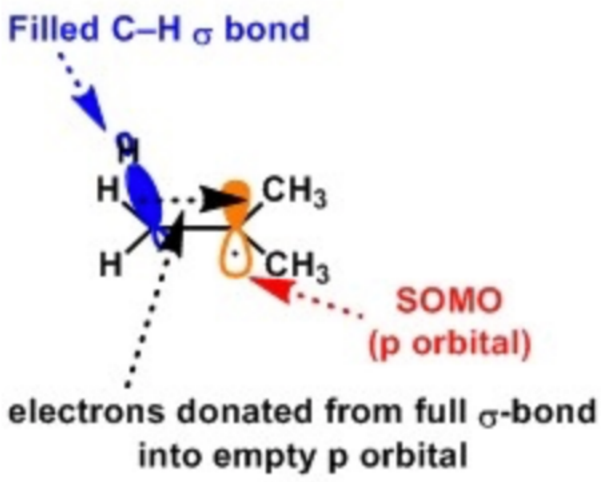
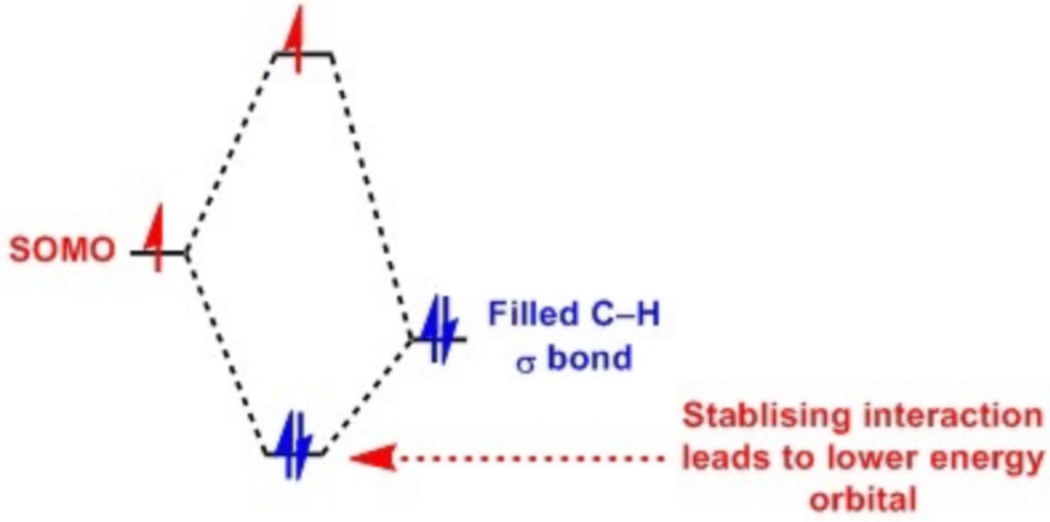
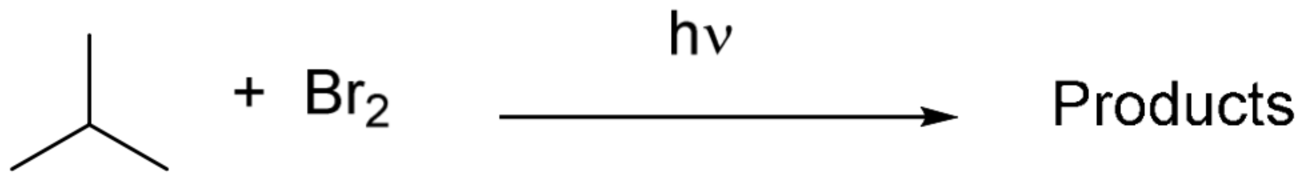


Hyperconjugation: Free radicals, like carbocations, are stabilized by substituents such as alkyl groups that can donate electrons to the unfilled orbital by hyperconjugation, where the unpaired electron, plus electrons in σ bonds that are β to the radical site, are delocalized. So more substituents, results in an increased stability.





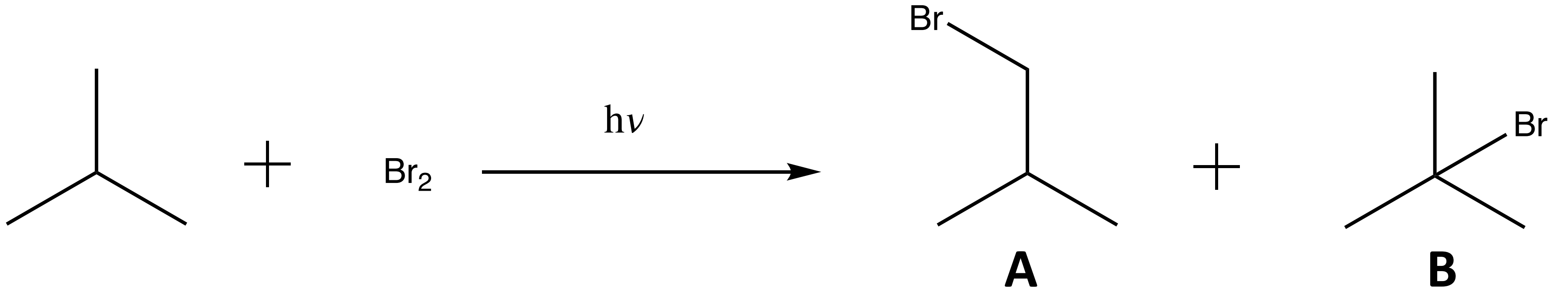
4) Consider the following reaction:



a) Draw the structure of all possible products, identify the main product and explain why it is the main product.
b) Write a mechanism for the reaction leading to the main product (A "reaction mechanism", more often simply "mechanism" shows, through the use of curly arrows, how bonds are broken and formed, in other words, is a way of showing the reaction pathway).

4) Consider the following reaction:

a) Draw the structure of all possible products, identify the main product and explain why it is the main product.

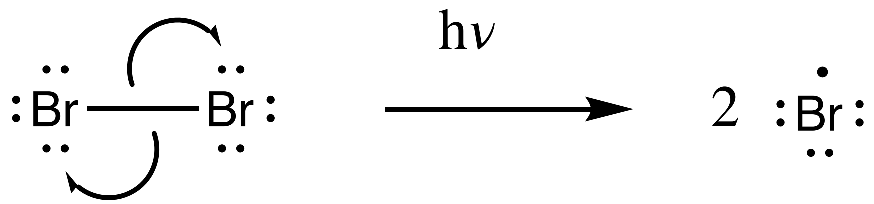


The rate determining step is the abstraction of the hydrogen atom. Abstracting a hydrogen is easier if the resulting radical is more stable. As tertiary radicals are more stable than primary radicals we find that mainly product **B** is formed.

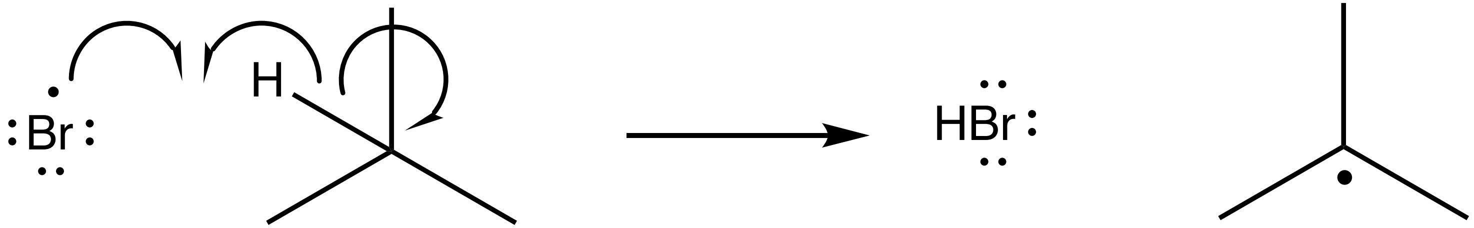
4) Consider the following reaction:

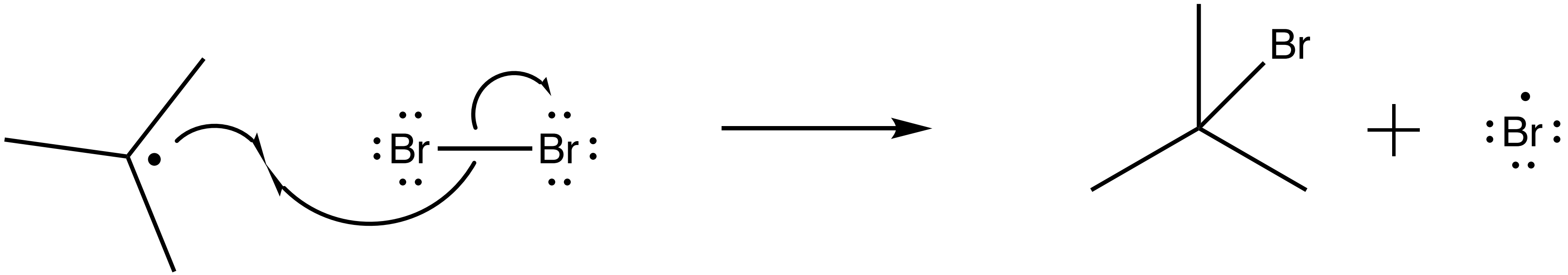
b) Write a mechanism for the reaction leading to the main product

Initiation



Propagation





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Termination